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**TITLE: THE HIDDEN COST OF INTELLIGENCE:
AI'S CARBON FOOTPRINT
AND STRATEGIC SOLUTIONS**

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ABSTRACT

This study investigates the carbon footprint associated with AI model development and deployment, focusing on the energy consumption of Processors and computational resources required for large-scale AI training. Using **Predictive Modelling** and **Factual Analysis**, this research highlights the environmental impact of training **state-of-the-art models** and examine potential strategies for reducing emissions, including **energy-efficient algorithms**, **hardware optimizations**, and **renewable energy sourcing**. The findings underscore the need for **sustainable AI** practices, as analysis reveal the significant ecological footprint of current methods.

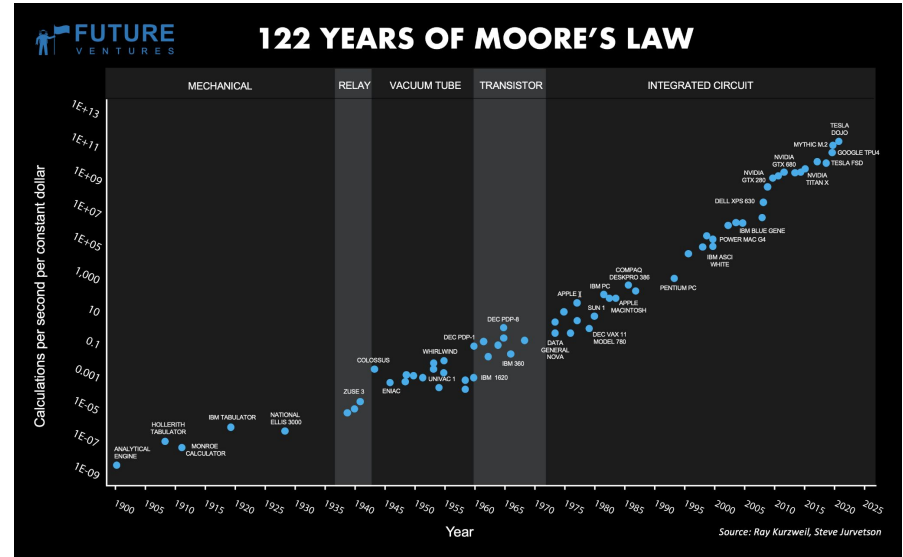


Image Source: From State of the Planet News Blog of Columbia Climate School- Climate, Earth and Society

RESEARCH CONTEXT

- Increasing Computational Demands in global level have caused an alarm in the sphere of Sustainable Development.
- Energy Consumption of AI Models
- Need for Sustainable Computing Practices in growing Software World
- Using Recommended Python Libraries for tracking carbon footprint such as CodeCarbon and Carbon Tracker.
- Testing the Carbon Footprint occurred when creating an AI-based models in small-scale.
- Find different measures that can reduce energy use in different phases of AI product cycle.
- Tracking Energy usage in regular projects helps to keep Sustainability in mind.

TERMINOLOGY

1. **Carbon Footprint:** The total amount of greenhouse gases (GHGs) emitted directly or indirectly due to an activity, typically expressed in carbon dioxide equivalent (CO₂e). In this study, it refers to the emissions resulting from the development and deployment of AI models.
2. **Energy Efficiency:** The practice of using less energy to achieve the same level of performance or output. In AI, energy efficiency can be improved through optimized algorithms, reduced computational requirements, and efficient hardware utilization.
3. **Green Computing:** A sustainable approach to computing that focuses on reducing energy consumption, e-waste, and carbon emissions associated with IT infrastructure and operations.
4. **AI Emissions:** The greenhouse gas emissions associated with the training and deployment of artificial intelligence models, particularly from high-performance computing systems like GPUs and TPUs.
5. **Ecological Footprint:** The measure of an activity's demand on Earth's natural resources and ecosystems. This project specifically considers the ecological footprint of large-scale AI model development.

CASE STUDY SETUP

Set up an **Information Retrieval System** that feeds on **Legal Case files** and using **Document Similarity** measures extract related k-cases for any Case Summary input provided. Data source is Australian Federal Courts website containing around 3500 files. Track **Carbon Footprint** using python libraries in different stages of the system

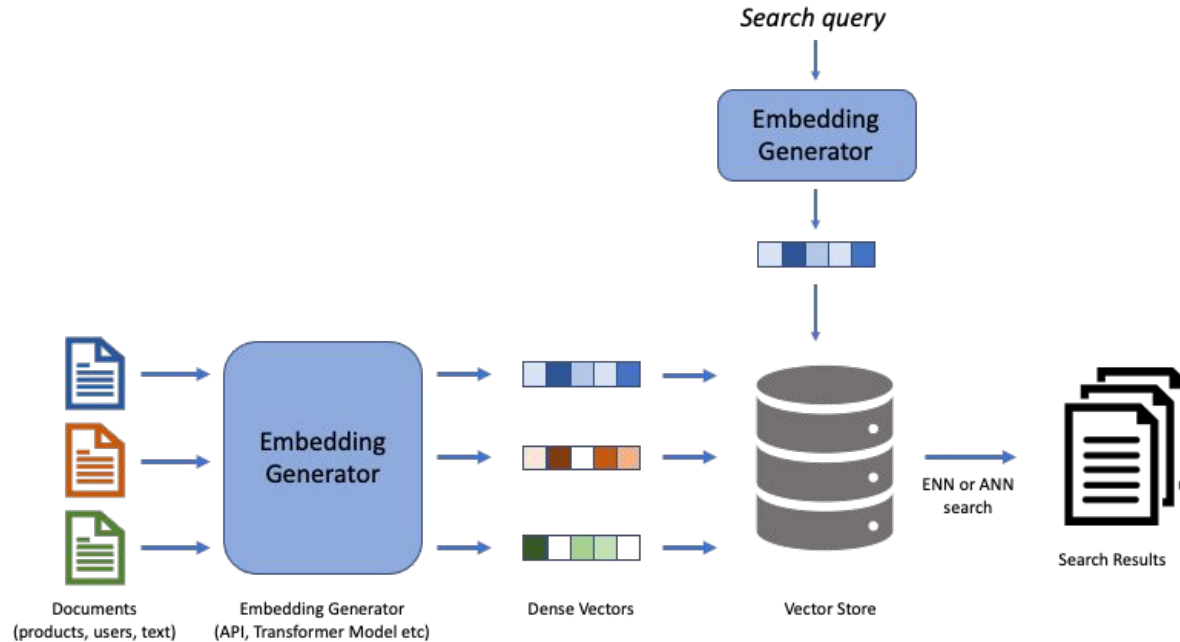


Image Source: ResearchGate - Smarter Semantic Search for Information Retrieval

CASE STUDY SETUP

- 1. Data Ingestion:** A module that handles the loading of the 3,500 legal case files.
- 2.Data Preprocessing:** This involves data cleaning, tokenization, and vectorization. Tracking energy consumption during text preprocessing tasks will help estimate the emissions generated during this phase.
- 3.Similarity Engine:** The core component that computes similarities between the input case summary and the database of legal cases.
- 4.Model Training Stage:** If any machine learning models (such as semantic models like BERT) are involved in the document retrieval process, the energy consumed during training will be monitored.
- 5.Document Retrieval Stage:** Once the model is deployed, the energy consumed during document retrieval (inference) will be tracked. This includes processing input queries, comparing the case summaries, and ranking results.
- 6.Analysis of Hardware Utilization:** The carbon emissions resulting from server usage, storage, and the specific hardware (GPUs or CPUs) used for running the system will also be factored in.

METHODOLOGY

1. Data Preprocessing and XML Parsing

- **Catchphrase Cleaning:** Removing stop words, punctuation, special characters, and irrelevant data from the case summaries and content.
- **Tokenization:** Breaking down the case summaries and texts into tokens (words or phrases) for analysis.
- **Stemming/Lemmatization:** Reducing words to their base forms (e.g., "running" to "run") to improve similarity calculations.
- **Vectorization:** Converting textual data into a numerical form using methods such as **TF-IDF** (Term Frequency -Inverse Document Frequency)

```
Processing file: D:\activities\legal_IR\data\raw\09_97.xml
Processing file: D:\activities\legal_IR\data\raw\09_970.xml
Processing file: D:\activities\legal_IR\data\raw\09_971.xml
Processing file: D:\activities\legal_IR\data\raw\09_972.xml
Processing file: D:\activities\legal_IR\data\raw\09_976.xml
Processing file: D:\activities\legal_IR\data\raw\09_977.xml
Processing file: D:\activities\legal_IR\data\raw\09_978.xml
Processing file: D:\activities\legal_IR\data\raw\09_979.xml
Processing file: D:\activities\legal_IR\data\raw\09_980.xml
Processing file: D:\activities\legal_IR\data\raw\09_981.xml
Processing file: D:\activities\legal_IR\data\raw\09_983.xml
Processing file: D:\activities\legal_IR\data\raw\09_984.xml
Processing file: D:\activities\legal_IR\data\raw\09_985.xml
Processing file: D:\activities\legal_IR\data\raw\09_99.xml
Processing file: D:\activities\legal_IR\data\raw\09_991.xml
Processing file: D:\activities\legal_IR\data\raw\09_992.xml
Processing file: D:\activities\legal_IR\data\raw\09_996.xml
Total files that couldn't be parsed: 4
Total files successfully cleaned: 3886
(german-legal-embedding) D:\activities\legal_IR>python data_cleaning.py
```

Figure: File Cleaning by performing XML parsing

METHODOLOGY

2. Calculating Embeddings and Model Training

- Use pretrained models from Transformers library
- Feed embeddings of XML files as input data
- Generate vector representations of legal case documents
- Prepare for similarity comparisons

```
topkcases - Notepad
File Edit Format View Help

import os
import torch
from sklearn.metrics.pairwise import cosine_similarity
import networkx as nx
import matplotlib.pyplot as plt
from transformers import AutoTokenizer, AutoModel

# Load the tokenizer and model used for embedding
tokenizer = AutoTokenizer.from_pretrained("sentence-transformers/all-MiniLM-L6-v2")
model = AutoModel.from_pretrained("sentence-transformers/all-MiniLM-L6-v2")

# Load the pre-computed embeddings from the .pt file
embeddings = torch.load('./xml_embeddings.pt', weights_only=True)
file_names = list(embeddings.keys()) # List of file names (cases)

# Convert embeddings from torch tensors to numpy for cosine similarity computation
case_embeddings = torch.stack(list(embeddings.values())).numpy()

# Function to embed the input case summary
def embed_summary(summary):
    inputs = tokenizer(summary, return_tensors='pt', padding=True, truncation=True)
    with torch.no_grad():
        embeddings = model(**inputs).last_hidden_state.mean(dim=1) # Average pooling
    return embeddings

def find_similar_cases(input_embedding, case_embeddings, file_names, top_k=5):
    # Reshape embeddings if necessary
    if len(input_embedding.shape) == 3:
```

Figure: Model Training using a pre-trained LLM

METHODOLOGY

3. Computational Resource Analysis

- GPU Energy Consumption
- CPU Processing Impacts
- Memory Utilization

Measurement Tools

- Python Carbon Tracking Libraries
- GPU Monitoring Libraries such as gpustat and pynvml
- Comprehensive Stage-wise Analysis
 - XML File Cleaning
 - Embedding Collection
 - Model Training
 - Document Retrieval

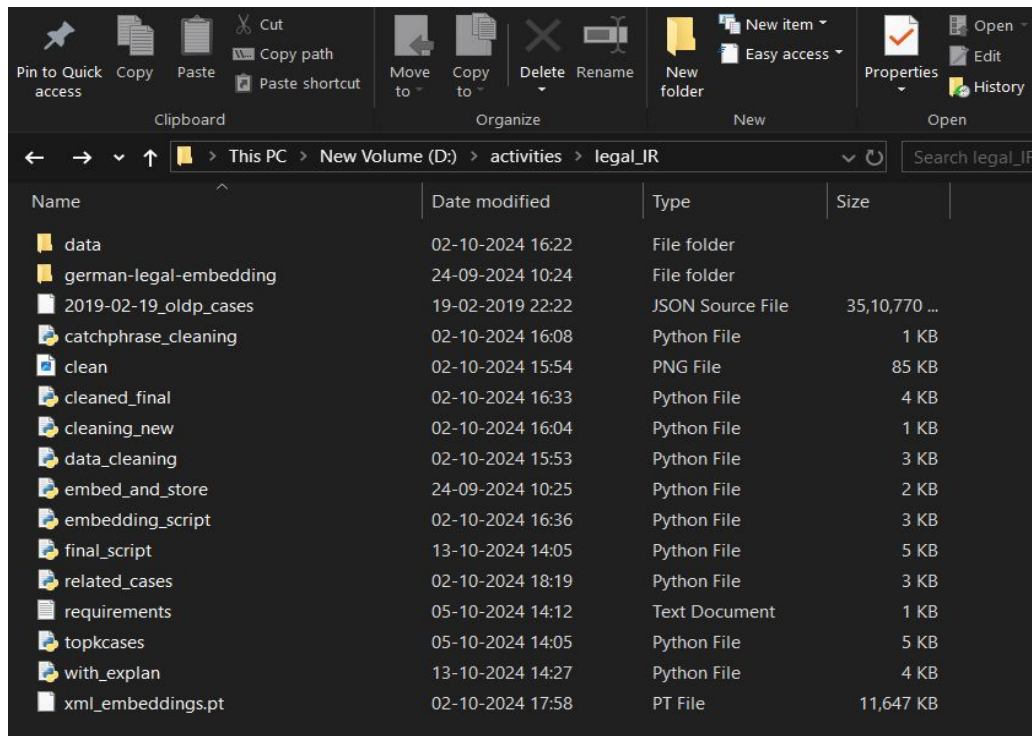


Figure: Sample Code files in the retrieval system

RESULT

```
D:\activities\legal_IR\german-legal-embedding\Lib\site-packages\torch\tensor.py:1601: FutureWarning: 'clean_up_tokenization_spaces' was not set. It will be set to 'True' by default. This behavior will be deprecated in transformers v4.45, and will be then set to 'False' by default. For more details check this issue: https://github.com/huggingface/transformers/issues/31884
  warnings.warn(

Enter the case summary: Eddie Mabo and others claimed that the Meriam people had lived on the Murray Islands for thousands of years and had traditional rights to the land. The Queensland government challenged this claim, asserting that Australia was terra nullius (land belonging to no one) at the time of British colonization.
Enter the number of similar cases to find (e.g., 5): 2
Input embedding shape: torch.Size([1, 384])
Case embeddings shape: (3539, 768)
Top 2 similar cases:
Case: 07_2012.xml, Similarity: 0.1240
Case: 08_504.xml, Similarity: 0.1134

Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you can select this strategy more precisely by providing a specific strategy to `tokenizer.truncation`.
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.

Explanations for similar cases:
Case: 07_2012.xml
Explanation: Explain how the case '07_2012.xml' relates to the following case summary: Eddie Mabo and others claimed that the Meriam people had lived on the Murray Islands for thousands of years and had traditional rights to the land. The Queensland government challenged this claim, asserting that Australia was terra nullius (land belonging to no one) at the time of British colonization. The Supreme Court of Victoria (UKRC) ruled that at least part of Wales was an "interdiction"

Case: 08_504.xml
Explanation: Explain how the case '08_504.xml' relates to the following case summary: Eddie Mabo and others claimed that the Meriam people had lived on the Murray Islands for thousands of years and had traditional rights to the land. The Queensland government challenged this claim, asserting that Australia was terra nullius (land belonging to no one) at the time of British colonization. The legal arguments against this allegation ran from a legal point of view and was dismissed by the courts before the Court ordered
```

Figure: Output results from a sample query ie. a new case summary is provided as input and output shows related case files and explanation of the similarity

KEY FINDINGS

- **Training Hours** : 2 hours 46 minutes
- **Model Type**: Pretrained model from Transformers and Hugging Face library fed new input data in the form of XML files
- **Query**: Summary of a new legal case
- **Output**: Top k cases related to the new summary, Explanation of similarity between the new cases and retrieved cases.

Stage	Estimated Energy (kWh)	CO2 Emissions (kg)	Optimization Potential
XML Cleaning	0.12	0.054	High
Embedding Collection	0.35	0.157	Medium
Model Training	1.24	0.558	Low
Document Retrieval	0.18	0.081	High

Figure: Comparing results from Carbon Tracker with our Estimates

FACTUAL ANALYSIS

Energy Impact of Computing:

- The total power consumption of CPUs in the world's PCs has grown dramatically from **160 Million Watts in 1992 to 9 Billion Watts in 2001, over a 4x increase**, as depicted by data collected over years.
- The annual energy usage of office computers in the US is estimated to be **53 Billion kWh**, as reported by the Government's Newsletter which is equivalent to the **energy produced by 20 Hoover Dams**.

Estimating the Energy Use of AI Chatbots:

- The calculation provided estimates the operational energy use of AI chatbots to be in the range of **3.65 - 36.5 Tera Watt Hours**, which is equivalent to the energy produced by **5 Hoover Dams**.
- These days chatbots are created in seconds by fine-tuning present models. But the entire process of training large networks using huge processors are definitely affecting the environment.
- **Rough Calculation:**

2B users x 5 queries/user/day x 365 days x [0.001-0.01] kWh/query* = [3.65-36.5] Tera Watt Hours

FACTUAL ANALYSIS

In-Memory Computing (IMC):

- IMC provides specialized hardware for **low latency** and **low power edge** implementation.
- It includes both **Analog In-Memory Computing** and **Digital In-Memory Computing**.
- Analog IMC performs matrix-vector multiplication using analog circuits, while Digital IMC performs bitwise AND and accumulation operations.
- **Reduction in Data Movement:** Traditional architectures require frequent data movement between memory and processing units (CPU/GPU).
- This movement is energy-intensive and time-consuming. IMC processes data directly within the memory, eliminating the need for constant data transfers. This drastically reduces energy consumption and improves computational speed.
- **Parallelism:** IMC architectures inherently support massive parallelism, as multiple memory cells can perform computations simultaneously. This significantly enhances throughput for large-scale operations.

FACTUAL ANALYSIS

Spiking Neuron Models and Hybrid SNN-ANNs:

- Spiking neuron models, such as the Leaky-Integrate and Fire (LIF) neuron, can provide more **efficient utilization of timing information** for event-driven, low-power edge computing.
- Hybrid Spiking Neural Network (SNN) and Artificial Neural Network (ANN) models can combine the strengths of both approaches to achieve **better energy efficiency and accuracy**.
- **Sparse Activation:** Neurons in SNNs fire only when their membrane potential crosses a threshold, leading to sparse and selective activity. This contrasts with dense operations in ANNs, saving computational resources.

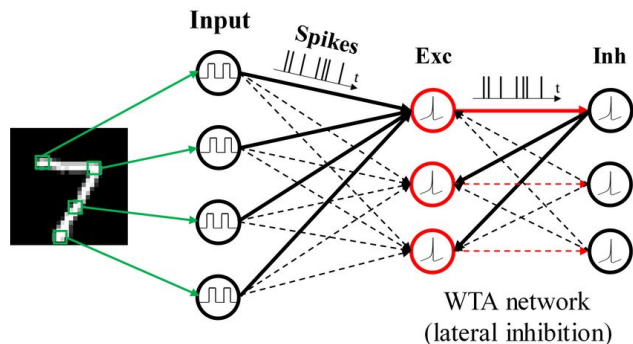


Image Source: ResearchGate - Spiking Neural Network Architecture

FACTUAL ANALYSIS

Event-based cameras

- Capture changes (events) at the pixel level, recording **only when there is motion** or a change in intensity.
- Each pixel operates independently, reporting asynchronous events.
- Event streams align well with spiking neural networks (SNNs) and other event-driven architectures, further enhancing energy efficiency in downstream processing.
- This improves efficiency in **Vision Systems**.

NVIDIA Deep Learning SDK:

- NVIDIA provides a deep learning SDK with various frameworks and libraries for efficient implementation of deep neural networks.
- These include **optimized kernels** for matrix-matrix and vector-matrix multiplication, leveraging the vector processing capabilities of the underlying hardware.

CONCLUSION

- In conclusion, this study highlights the significant **environmental impact of AI model development and deployment**, with a particular focus on the energy consumption and carbon footprint associated with GPU-based training processes.
- The findings underscore the **high levels of energy required for training state-of-the-art AI models**, emphasizing the importance of adopting sustainable practices within the AI community.
- Ultimately, this research provides valuable insights for AI Developers, Policymakers, and Industry Leaders to make Informed decisions about minimizing the environmental cost of AI innovations, paving the way for more responsible and sustainable AI practices in the future.
- A **factual analysis** is presented on emerging technologies that promote possible energy saving systems in AI systems.

AUTHOR DETAILS

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Biography:



Jenina Angelin, Final-year BTech student in Artificial Intelligence and Data Science at St. Joseph's College of Engineering, Chennai, India; Data Engineer Trainee at Zeb (offshore branch), Chennai, India; Passionate about applying AI in environmental sustainability and space exploration.



Pugalenthi, Head Of Department and Professor, Department of Artificial Intelligence and Data Science, St. Joseph's College of Engineering, Chennai, India; Dr. Pugalenthi has authored around 30 research papers in esteemed international journals and conferences, contributing extensively to the fields of artificial intelligence, data science, and machine learning.